

Linear Independence

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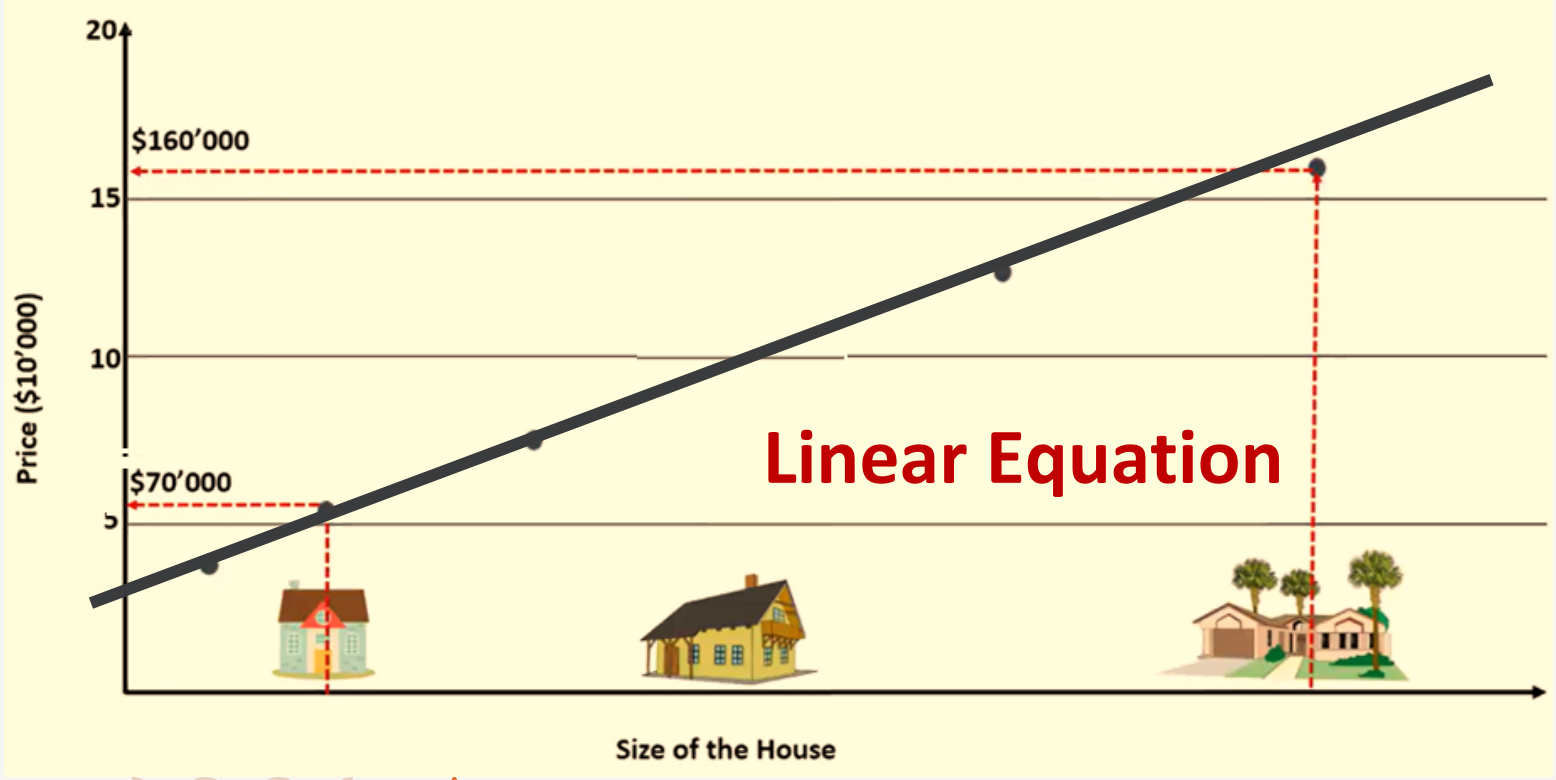
01

Introduction

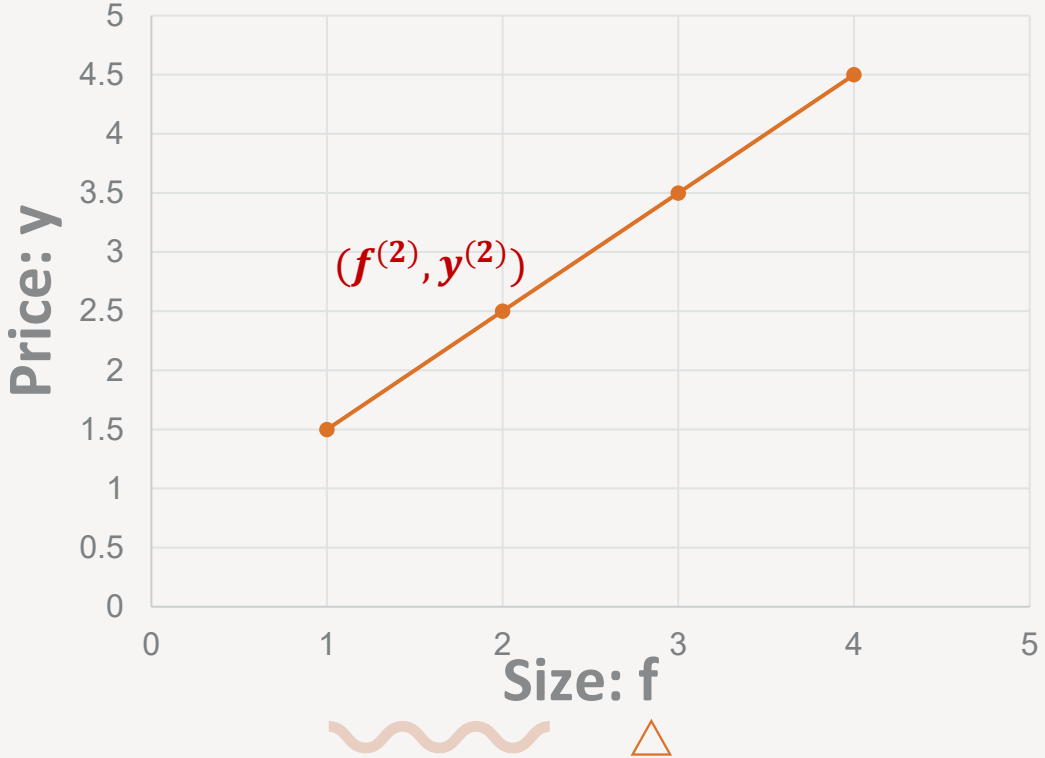
Price Problem



Price Problem



Linear Equation with offset



$$y = af + c$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$

House Features

- #Room
- Size
- #Bedroom
- Age
- Address features: Street, Alley, …
- Size of part1, part2, part3, part4
- Floors
- #Bathrooms
- …

Which features are dependent on others?

02

Linear Independence

Linear Independence (Algebra)

Definition

Dependent

- For at least one $\lambda \neq 0$ $\quad 0 = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n, \quad \lambda \in \mathbb{R}$
- A set of vectors is dependent if at least one vector in the set can be expressed as a linear weighted combination of the other vectors in that set.

Linear Independence (Algebra)

Definition

Independent

- Only when all $\lambda_i = 0$ $\qquad 0 = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n, \qquad \lambda \in \mathbb{R}$
- No vector in the set is a linear combination of the others (has only the trivial solution)

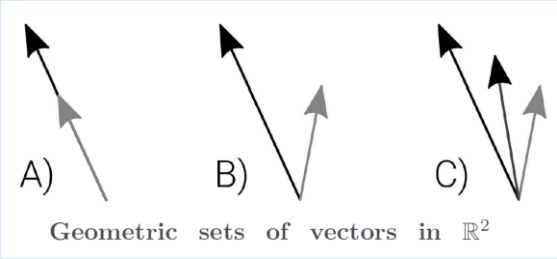
Linear Independence (Geometry)

Definition

A set of vectors is linear independent if the subspace dimensionality (its span) equals the number of vectors.

Example

- Are these vectors linearly independent?



Geometric sets of vectors in $\mathbb{R}^2$

Example

Example

- Let $v_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}, v_2 = \begin{bmatrix}4\\5\\6\end{bmatrix}, and\ v_3 = \begin{bmatrix}2\\1\\0\end{bmatrix}.$

- a) $v_1 = \begin{bmatrix}3\\1\end{bmatrix}, v_2 = \begin{bmatrix}6\\2\end{bmatrix}$

b) $v_1 = \begin{bmatrix}3\\2\end{bmatrix}, v_2 = \begin{bmatrix}6\\2\end{bmatrix}$

Example Solution!

- $v_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}, v_2 = \begin{bmatrix}4\\5\\6\end{bmatrix}, and\ v_3 = \begin{bmatrix}2\\1\\0\end{bmatrix}$:

$$-2v_1 + v_2 = \begin{bmatrix}-2\\-4\\-6\end{bmatrix} + \begin{bmatrix}4\\5\\6\end{bmatrix} = \begin{bmatrix}2\\1\\0\end{bmatrix} = v_3$$

=> Three vectors are dependent!

- $v_1 = \begin{bmatrix}3\\1\end{bmatrix}, v_2 = \begin{bmatrix}6\\2\end{bmatrix}$:

$$v_2 = 2v_1$$

=> They are linearly dependent!

- $v_1 = \begin{bmatrix}3\\1\end{bmatrix}, v_2 = \begin{bmatrix}6\\2\end{bmatrix}$:

There is no scalar λ with $v_2 = \lambda v_1$. Hence they are **linearly independent!**

Properties

Theorem 1

Any set of vectors that contains the zeros vector is guaranteed to be linearly dependent.

Properties

Proof

By definition, vectors are linearly independent only if

$$c_1 v_1 + \cdots + c_k v_k + c_{k+1} 0 = 0$$

implies that all coefficients are zero.

But if we choose $c_1 = \cdots = c_k = 0,\ c_{k+1} = 1$, the equation still holds because $1 \cdot 0 = 0$.

Thus, a non-trivial solution exists, proving the set is linearly dependent.

Characterization of Linearly Dependent sets

Theorem 2

An indexed set $S = \{v_1, \dots, v_n\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$, then **some** v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \dots, v_{j-1}$.

Proof

Notes!!!

- Does not say that every vector
- Does not say that every vector in a linearly dependent set is a linear combination of the preceding vectors. A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

Characterization of Linearly Dependent sets

Proof

Let j be the largest subscript for which $c_j \neq 0$. If $j = 1$, then $c_1 v_1 = 0$, which is impossible because $v_1 \neq 0$. So $j > 1$ and

$$c_1 v_1 + \cdots + c_j v_j + 0 v_{j+1} + 0 v_n = 0$$

$$c_j v_j = -c_1 v_1 - \cdots - c_{j-1} v_{j-1}$$

$$v_j = \left(-\frac{c_1}{c_j}\right) v_1 + \cdots + \left(-\frac{c_{j-1}}{c_j}\right) v_{j-1}$$

Characterization of Linearly Dependent sets

Proof

If some v_j in S equals a linear combination of the other vectors, then v_j can be subtracted from both sides of the equation, Producing a linear dependence relation with a nonzero weight (-1) on v_j. [For instance, if $v_1 = c_2v_2 + c_3v_3$, then $0 = (-1)v_1 + c_2v_2 + c_3v_3 + 0v_4 + \cdots + 0v_n$.] Thus S is linearly dependent.

Conversely, suppose S is linearly dependent. If v_1 is zero, then it is a (trivial) linear combination of the other vectors in S. Otherwise, $v_1 \neq 0$, and there exist weights $c_1, \ldots, c_n$ not all zero, such that

$$c_1v_1 + c_2v_2 + \cdots + c_pv_n = 0$$

Properties

Theorem 3

- Any set of $\mathrm{p} > n$ vectors in $\mathbb{R}^n$ is necessarily dependent.
- Any set of $\mathrm{p} \leq n$ vectors in $\mathbb{R}^n$ could be linearly independent.

Proof

$$\begin{matrix} & p \\ n & \end{matrix}\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}$$

Exercise

Example

a. $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$

b. $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 8 \end{bmatrix}$

c. $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ -9 \\ 15 \end{bmatrix}$

Exercise Solution!

- $\begin{bmatrix}1\\7\\6\end{bmatrix}, \begin{bmatrix}2\\0\\9\end{bmatrix}, \begin{bmatrix}3\\1\\5\end{bmatrix}, \begin{bmatrix}4\\1\\8\end{bmatrix}$: $p > n \Rightarrow$ vectors are dependent!
- $\begin{bmatrix}2\\3\\5\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\8\end{bmatrix}$: contains zero ⇒ vectors are linearly dependent!
- $\begin{bmatrix}-2\\4\\6\\10\end{bmatrix}, \begin{bmatrix}3\\-6\\-9\\15\end{bmatrix}$: one vector is not a scalar multiple of the other ⇒ they are linearly independent!

Linear Dependent Properties

- Suppose vectors $v_1, \dots, v_n$ are linearly dependent:

$$c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = 0$$

with $c_1 \neq 0$. Then:

$$span\{v_1, \dots, v_n\} = span\{v_2, \dots, v_n\}$$

- When we write a vector space as the space of a list of vectors, we would like that list to be as short as possible. This can be achieved by iterating.

Linear combinations of linearly independent vectors

Theorem 4

Suppose x is linear combination of linearly independent vectors $v_1, \dots, v_n$:

$$x = \beta_1 v_1 + \cdots + \beta_n v_n$$

The coefficients $\beta_1, \dots, \beta_n$ are unique.

Linear combinations of linearly independent vectors

Proof

Assume there are two representations of x:

$$x = \beta_1 v_1 + \cdots + \beta_n v_n = \gamma_1 v_1 + \cdots + \gamma_n v_n$$

Subtract one equation from the other:

$$(\beta_1 - \gamma_1) v_1 + \cdots + (\beta_n - \gamma_n) v_n = 0$$

Since the vectors are linearly independent, the only solution is:

$$(\beta_1 - \gamma_1) = \cdots = (\beta_n - \gamma_n) = 0$$

Hence, $\beta_i = \gamma_i$ for all i. Therefore, the coefficients are unique. ✅

Conclusion

- Step 1: Count the number of vectors (call that number p) in the set and compare to n in $\mathbb{R}^n$. As mentioned earlier, if $\mathrm{p} > n$, then the set is necessarily dependent. If $\mathrm{p} \leq n$ then you have to move on to step 2.
- Step 2: Check for a vector of all zeros. Any set that contains the zeros vector is a dependent set.
- The rank of a matrix is the estimate of the number of linearly independent rows or columns in a matrix.

03

Functions Linearly Independent

Functions Linearly Independent

- Let f(t) and g(t) be differentiable functions. Then they are called linearly dependent if there are nonzero constants c_1 and c_2 with

$$c_1 f(t) + c_2 g(t) = 0$$

for all t. Otherwise, they are called linearly independent.

Example

Linearly dependent or independent?

- Functions $f(t) = 2\sin^2 t$ and $g(t) = 1 - cos^2 t$
- Functions $\{\sin^2 x, \cos^2 x, \cos(2x)\} \subset \mathcal{F}$

Example Solution!

- Functions $f(t) = 2\sin^2 t$ and $g(t) = 1 - cos^2 t$:

$$f(t) - 2g(t) = 2\sin^2 t - 2 + 2\cos^2 t = 2 - 2 = 0 \Rightarrow \text{They are dependent!}$$

- Functions $\{\sin^2 x, \cos^2 x, \cos(2x)\} \subset \mathcal{F}$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$\Rightarrow 1\sin^2 x - 1\cos^2 x + 1\cos(2x) = -\cos(2x) + \cos(2x) = 0$$

$\Rightarrow$ They are linearly dependent!

04

Polynomials Linearly Independent

Vector Space of Polynomials

Example

Are $(1-x), (1+x), x^2$ linearly independent?

Example Solution!

$$c_1(1-x) + c_2(1+x) + c_3x^2 = 0$$

$$\Rightarrow (c_1 + c_2) + (-c_1 + c_2)x + c_3x^2 = 0$$

$$\Rightarrow c_1 + c_2 = 0, -c_1 + c_2 = 0, c_3 = 0$$

$$\Rightarrow c_1 = c_2 = c_3 = 0$$

⇒They are linearly independent!

Resources

- Page 97 LINEAR ALGEBRA: Theory, Intuition, Code
- Page 213: David Cherney,
- Page 54: Linear Algebra and Optimization for Machine Learning

Properties

- The vectors coming from the vector form of the solution of a matrix equation Ax = 0 are linearly independent

Example

- Vectors related to x_2 and x_3 are linear independent.
- Columns of A related to to x_2 and x_3 are linear dependent.
- Span$\{A_1, A_2, A_3\} = Span\{A_1\}$

$$\mathrm{A} = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -4 \end{bmatrix} \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

Properties

Important

- If a collection of vectors is linearly dependent, then any superset of it is linearly dependent.
- Any nonempty subset of a linearly independent collection of vectors is linearly independent.